

# M. Ahmad Siddiqui

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## PROFESSIONAL SUMMARY

Data Scientist and ML Engineer with hands-on experience in deep learning, NLP, computer vision, and edge AI deployment. Built end-to-end systems spanning RAG pipelines, LLM fine-tuning (LoRA/QLoRA), on-device inference with ONNX Runtime and llama.cpp, and real-time computer vision. Skilled at taking projects from research to working implementation.

## EXPERIENCE

### REZINX

Contractual ML Engineer

Remote

Aug 2025 – Present

- Fine-tuned language models using LoRA for downstream production use cases.
- Built a model and systems monitoring dashboard using Grafana.
- Evaluated multiple embedding models for a job classification task.
- Built a recommendation engine using Superlinked.
- Developed a computer vision hazard detection system to identify fires, knives, and stairs, and assess their proximity to children.

### Ittehad Chemicals

Intern

Lahore, Punjab, Pakistan

Jun 2026 – Jul 2026

- Assisted in daily operational tasks, ensuring adherence to safety protocols and standards.
- Collaborated with team members to streamline production processes and improve efficiency.
- Collaborated with team members to streamline inventory management processes.

### Arch Technologies

Intern

Islamabad, Pakistan (Remote)

May 2025 – June 2025

- Developed machine learning models to analyze large datasets and identify patterns.
- Assisted in data preprocessing and feature engineering for enhanced model accuracy.
- Conducted experiments to evaluate model effectiveness and recommend adjustments.

## EDUCATION

**Khwaja Fareed University of Engineering and Information & Technology**

Aug 2022 – May 2026

Bachelor of Science in Data Science | CGPA: 3.35/4.00

## PROJECTS

### QuestionPrep: AI-Based Interview Simulation Platform (FYP)

- Built a full-stack AI interview platform using FastAPI, llama.cpp (LFM-1.2B GGUF), ChromaDB with BGE embeddings, and a three-stage NLI scoring pipeline (fuzzy match → bi-encoder → DeBERTa cross-encoder).
- Implemented real-time stress detection in-browser via a TCN model exported to ONNX Runtime Web, using a custom JavaScript DSP pipeline (FFT, mel filterbank, DCT-II) for feature extraction with no server round-trip.

### Hybrid Profanity Detection & Rewriting System (NLP)

- Engineered a two-stage inference pipeline combining a LoRA-fine-tuned DistilBERT classifier (95.67% accuracy) with a QLoRA-fine-tuned LFM-1.2B LLM for context-aware rewriting.
- Implemented dynamic adapter switching between classification and generation modes; documented generation collapse as an intrinsic post-classification fine-tuning phenomenon.

### Cyber Attack Detection System (Machine Learning)

- Implemented a multi-class network intrusion classifier on the Edge-IIoT dataset, reaching 94.67% accuracy.
- Applied ensemble techniques (stacking/boosting) with feature engineering to improve detection across attack categories.

## SKILLS

**Programming Languages:** Python, SQL

**ML & DL Frameworks:** PyTorch, scikit-learn, XGBoost, ONNX Runtime, llama.cpp

**NLP & LLMs:** Transformers (HuggingFace), LoRA, QLoRA, RAG, text classification, embeddings, LLM fine-tuning

**Computer Vision:** EfficientNet, YOLOv8, transfer learning, object detection, image classification, OpenCV

**Data & Backends:** FastAPI, ChromaDB, SQLite, Superlinked, vector search, Docker, REST APIs

**MLOps & Tools:** Grafana, Git, Bash, Jupyter